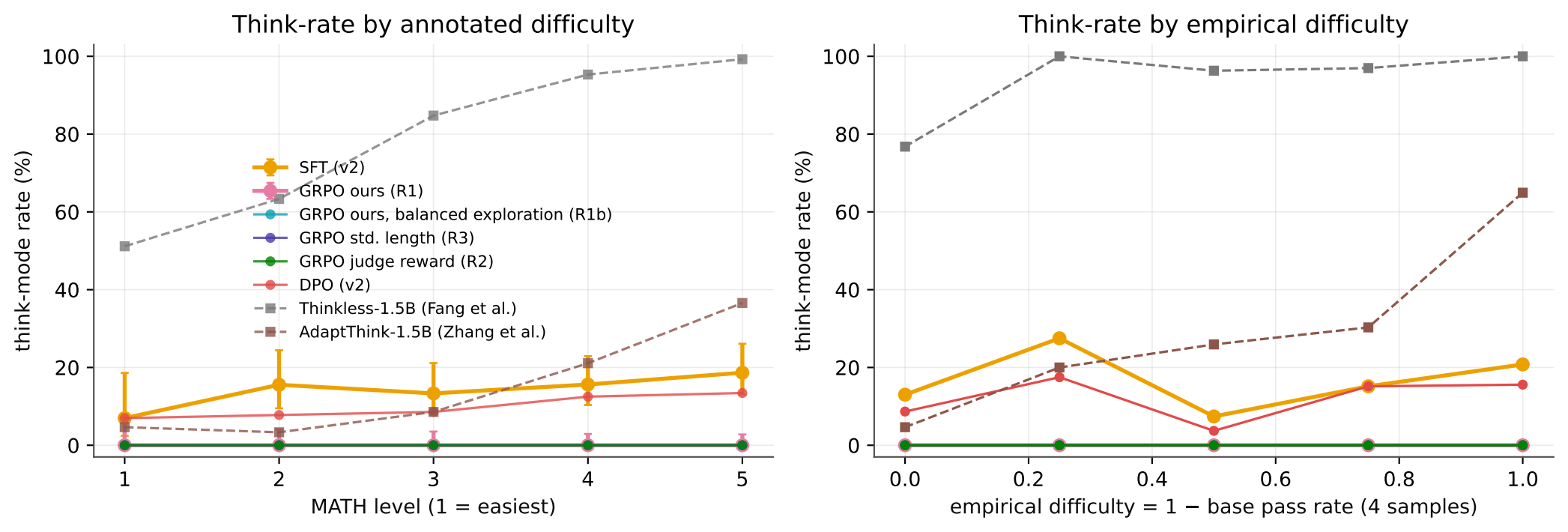
Think-rate by annotated difficulty
think-mode rate (%)
MATH level (1 = easiest)
SFT (v2)
GRPO ours (R1)
GRPO ours, balanced exploration (R1b)
GRPO std. length (R3)
GRPO judge reward (R2)
DPO (v2)
Thinkless-1.5B (Fang et al.)
AdaptThink-1.5B (Zhang et al.)
Think-rate by empirical difficulty
think-mode rate (%)
empirical difficulty = 1 − base pass rate (4 samples)